

13.10 Classifying Quadrilaterals – Part 3

Lesson Introduction

 Benchmark	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.		 Learning Target	I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.
 Benchmark Coherence	Building On		Working Toward	
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through the given point.		N/A	
 Essential Questions	- Which quadrilateral properties can be used to classify the shape? - How can the shape be determined by the slope, side lengths, and/or diagonal lengths? - What verifications best fit the information found per card given in the activity?			
 Lesson Narrative	Students complete an activity using stations where they apply coordinate geometry to determine the type of quadrilateral the information represents.			
 Component Summary	13.10.1 Warm-Up (~5 Min)	Would You Rather? (SE p. 340)	13.10.2 Activity (40 Min)	Stations (SE p. 340)
	13.10.3 Wrap-Up (~5 Min)	Students summarize their learning (SE p. 344)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		13.10.2 Station Activity Cards (Optional) Calculators (Optional) Graphing Utility or Graph Paper	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with remembering the properties of all the quadrilaterals. - Students could plot the vertices incorrectly, and lead to wrong calculations.	- Consider how best to organize the activity for your students. The activity can be completed by students individually, with partners, or in a small group. - If using small groups, consider making one of the stations a small group for students who need further support to work with you. - A calculator is not necessary for 13.10.2 Activity because students can leave values in radical form. Consider making 13.10.2 no calculator, and then allowing a calculator in the activity in 13.11.2
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Students can complete the 13.10.2 Activity with a partner or small group. Students can form their own group or be placed by the teacher. Students can be dealt one card or several as a group. Each group can read the cards to each other, and think about justifications of the problem while working it out. Students then can share with each other or the class as to what justifications they found and give reasoning including measurements, shape sketches, and properties to use in order to classify the quadrilateral.	MA.K12.MTR.2.1: Multiple Representations Students are completing an activity for 13.10.2, where they are asked to classify and justify the quadrilateral that is created with the information calculated and found per card. Students will be given multiple representations of each quadrilateral, and are asked to determine different information to classify the given quadrilateral, or to identify the missing coordinate for the quadrilateral given.
 Differentiate Instruction	For struggling learners, consider handing the students the cards one at a time to complete. This will help keep students from getting overwhelmed, and help to focus on one problem at a time. Consider offering graphing paper or a graphing utility to assist visual learners. Auditory learners will benefit from working in a group to engage in discussions about classifying each quadrilateral. Placing the task cards around the room will help kinesthetic learners engage in the lesson.	
Lesson Notes		

13.10.1 Warm-Up: Would You Rather?

Component Overview: Students are asked to give their opinion on if they would rather go to the movies or watch tv shows.



13.10.1 Warm-Up: Would You Rather?

1. Would you rather ...

Give up watching
movies?

OR

Give up watching
TV shows?

Answers will vary.

2. Explain your choice.

Answers will vary.



- Why would the choice be movies over watching tv shows?
- Why would the choice be watching tv shows over choosing to go to the movies?
- What were the thoughts that came to mind as the question was read?

13.10.2 Activity: Stations

Component Overview: Students will be given cards that act as stations, and use the information on a card to classify the quadrilateral that is represented.



13.10.2 Activity: Stations

Complete the table for each question card.

Question	Work		Answer
1	Slopes	Slopes of $\overline{RB}$ and $\overline{KM} = 1$ Slopes of $\overline{BM}$ and $\overline{RK} = -1$	Properties Opposite sides are parallel. Adjacent sides are perpendicular.
	Side Lengths	$RB = \sqrt{18} = 4.243$ $BM = \sqrt{18} = 4.243$ $KM = \sqrt{18} = 4.243$ $RK = \sqrt{18} = 4.243$	All four sides are congruent. Diagonals are congruent.
	Diagonal Lengths	$RM = 6$ $BK = 6$	Quadrilateral Square

Question	Work		Answer
2	Slopes	Slopes of $\overline{NH}$ and $\overline{DP}$ are undefined. Slopes of $\overline{ND}$ and $\overline{HP} = \frac{-3}{2}$	Properties Opposites sides are parallel and congruent.
	Side Lengths	$NH = 3$ $DP = 3$ $ND = \sqrt{13} = 3.606$ $HP = \sqrt{13} = 3.606$	
	Diagonal Lengths	$NP = \sqrt{40} = 6.325$ $HD = 2$	Quadrilateral Parallelogram



There are 8 cards that have information on them for the student to use to classify the quadrilateral that the information represents. Students record their work and answer in the workbook. The activity offers flexibility, so determine what works best for your students. If utilizing all 8 cards, then divide your class into 8 or 9 groups. The ninth group could be where students work with you to either have you check their work to that point, or to work on common misconceptions you have noticed. Set up the groups and the order so that only those students who need further support will visit your station. Eight cards are provided so that you have the flexibility to use some or all of them. Consider using a timer to note when it is time for students to rotate. Additional questions are provided for you to consider.



Consider using these suggestions for the Stations Activity:

- Consider providing students with graph paper or a graphing utility, if needed.
- Lower-level students could only be given a few cards at a time to complete.
- Middle-level students could be given about 5-6 cards to complete for the class time.
- Upper-level students could be given all 8 cards to complete.
- The teacher can group students according to peer partners or small groups, depending on the teacher knowing how the students work best.

13.10.2 Activity: Stations (continued)

Question	Work		Answer
3	Length of $\overline{WH}$ and $\overline{HC}$	$WH = \sqrt{5} = 2.236$	$(-10,1)$
		$HC = \sqrt{10} = 3.162$	
	Slopes of $\overline{WH}$ and $\overline{HC}$	Slope of $WH = \frac{1}{2}$ Slope of $HC = -\frac{1}{3}$	Verification $WS = \sqrt{10} = 3.162$ $(-10,1)$ would be point S because $\overline{WH}$ has the same slope as $\overline{CS}$ and $\overline{WS}$ is congruent to $\overline{HC}$.
	Slope of $\overline{CS}$ Using Each Point S	$(-6, -2) = 1$ $(-10,1) = \frac{1}{2}$ $(-12,3) = \frac{1}{8}$	



- How can the lengths and slopes help you determine the coordinates for Point S ?
- If quadrilateral $WHCS$ was just a trapezoid, how would your thought process change?

Question	Work		Answer
4	Slopes	Slopes of $\overline{MT}$ and $\overline{FP} = 1$ Slopes of $\overline{MP}$ and $\overline{TF} = -\frac{3}{4}$	Properties Opposite sides are parallel and congruent.
		$MT = \sqrt{18} = 4.243$ $FP = \sqrt{18} = 4.243$ $MP = \sqrt{25} = 5$ $TF = \sqrt{25} = 5$	
	Diagonal Lengths	$MF = \sqrt{37} = 6.083$ $TP = 7$	Who is correct? Marissa



- What are the methods that was taken on card #4 to determine who was correct and who was not?

13.10.2 Activity: Stations (continued)

Question	Work		Answer
5	Slopes	<i>Slope of $\overline{KR} = 4$</i> <i>Slope of $\overline{KC} = \frac{2}{5}$</i> <i>Slope of $\overline{RN} = \frac{1}{5}$</i> <i>Slope of $\overline{NC} = \frac{4}{2}$</i>	Properties <i>Exactly two pairs of adjacent sides are congruent.</i>
		<i>$KR = \sqrt{17} = 4.123$</i> <i>$KC = \sqrt{29} = 5.385$</i> <i>$RN = \sqrt{17} = 4.123$</i> <i>$CN = \sqrt{29} = 5.385$</i>	
	Side Lengths		
	Diagonal Lengths	<i>$KN = \sqrt{18} = 4.423$</i> <i>$RC = \sqrt{72} = 8.485$</i>	Quadrilateral <i>Kite</i>



- When you drew the quadrilateral on the coordinate grid, what quadrilateral did you think it was?
- What information about the quadrilateral, if any, did the slopes provide?

Question	Work		Answer
6	Slopes	<i>Slopes of $\overline{BR}$ and $\overline{WH} = -2$</i> <i>Slopes of $\overline{WB}$ and $\overline{HR} = \frac{3}{5}$</i> <i>Slopes of $\overline{TF}$ and $\overline{NS} = \frac{1}{2}$</i> <i>Slopes of $\overline{TN}$ and $\overline{FS} = -2$</i>	Which quadrilateral is a rectangle? <i>Quadrilateral TFSN is a rectangle</i>
		<i>$BR = \sqrt{20} = 4.472$</i> <i>$WH = \sqrt{20} = 4.472$</i> <i>$WB = \sqrt{34} = 5.831$</i> <i>$HR = \sqrt{34} = 5.831$</i>	
	Side Lengths	<i>$TF = \sqrt{20} = 4.472$</i> <i>$NS = \sqrt{20} = 4.472$</i> <i>$TN = \sqrt{45} = 6.708$</i> <i>$FS = \sqrt{45} = 6.708$</i>	Properties <i>Opposite sides are parallel and adjacent sides are perpendicular.</i> <i>Opposite sides are congruent.</i> <i>Diagonals are congruent.</i>
	Diagonal Lengths	<i>$WR = \sqrt{50} = 7.071$</i> <i>$BH = \sqrt{58} = 7.616$</i> <i>$TS = \sqrt{65} = 8.062$</i> <i>$FN = \sqrt{65} = 8.062$</i>	



- Could you easily determine which quadrilateral was the rectangle from the graph?
- What information about the quadrilateral, if any, did the diagonals provide?

13.10.2 Activity: Stations (continued)

Question	Work	Answer
7	Slopes <i>Slopes of $\overline{XY}$ and $\overline{WV} = 2$ Slopes of $\overline{YV}$ and $\overline{XW} = \frac{1}{2}$</i>	Why is $VWXY$ not a square? <i>The slopes of adjacent sides are not opposite reciprocals to indicate perpendicular lines.</i>
	Side Lengths <i>$XY = \sqrt{20} = 4.472$ $WV = \sqrt{20} = 4.472$ $YV = \sqrt{20} = 4.472$ $XW = \sqrt{20} = 4.472$</i>	<i>All sides are congruent proving that the quadrilateral is a rhombus.</i>
	Diagonal Lengths <i>$YW = \sqrt{8} = 2.828$ $XV = \sqrt{72} = 8.485$</i>	<i>Diagonals are not congruent proving that the rhombus is not a square.</i>



- What qualities does a rhombus have that a square does not?
- How is a square different than a rhombus?

13.10.2 Activity: Stations (continued)

Question	Work		Answer
8	Slope and Length of $\overline{KD}$	$m = 1$ $KD = \sqrt{18}$ $= 4.243$	<i>(7,3) would be the point S.</i>
	Slope of $\overline{MS}$ Using Each Point S	$(4,5) = -\frac{3}{2}$ $(6,3) = -\frac{5}{4}$ $(7,3) = -1$	<i>The slope of $\overline{KD}$ is 1 and the slope of $\overline{MS}$ is -1 making the diagonals perpendicular.</i> $MD = 3$ $MK = 3$ $DS = \sqrt{29}$ $KS = \sqrt{29}$ <i>A kite has exactly two pairs of adjacent sides congruent.</i>



- Could you easily determine where point S should be located using the graph?
- Which, if any, information was most useful to help determine which point S was correct?
 - Slope of $\overline{KD}$
 - Length of $\overline{KD}$



Consider challenging students who finish early by having them to create their own activity card to share with another person in their group/class to answer.

13.10.3 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



13.10.3 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

Answers will vary.